MINISTRY OF EDUCATION AND TRAINING
CAN THO UNIVERSITY

SUMMARY REPORT RESEARCH PROJECT

- Project code: THS2026-68
- Project title: Development of an Edge AI-based System for Multimodal Analysis and Clustering of Flood Rescue Events
- Project period: May–October 2026
- Total cost: 15,000,000 VND
- Project leader: Lê Thị Ngọc Ảnh (0376932994)
- Project members: Nguyễn Thanh Trọng, Cao Tường Hưng, Nguyễn Như Quỳnh, Ngô Hưng Thịnh.
- Advisor: TS. Nguyễn Thanh Khoa (ntkhoa@ctu.edu.vn)
- Necessity of the project: Duplicate, unlocated or delayed rescue reports hinder coordination under intermittent connectivity.
- Objectives: Develop a prototype combining edge image inference, compact transmission and map-based event clustering.
- Methodology: Train MobileNetV3-Large and evaluate image classification, clustering, ranking and dispatch separately using saved artifacts and simulation datasets.
- Project activities: Image AI, structured urgency, persistent queues, adaptive sending, synchronization and dashboard development; conditional edge and copy-invariance audits.
- Research results: The original seed-1024 model achieves 75.78% accuracy and 74.26% macro-F1 on 256 test images in CPU evaluation, with three critical errors. Runtime conversion, quantization and pruning were evaluated separately; quantization reduced model size but also accuracy. Phone benchmarks cover 32 configurations on four devices. A Flutter app, FastAPI/SQLite backend and dashboard were developed.
- Research new finding: RQ1–RQ3 reveal conditional safeguards and limitations without establishing general superiority or field rescue benefits.
- Application potentials: Research and workflow demonstration. Text modeling, urgency integration, mobile provenance, repeated device/network/SMS tests and expert validation remain necessary.